Sokoine University of Agriculture (SUA)
Department of Agricultural Engineering
Electronics and Precision Agriculture Lab (EPAL)

AGE 418
Sensors and Controls in Precision Agriculture

Lecture 14 Compendium:
Economic and Environmental Impacts
The Triple Bottom Line of Precision Ag

Instructor: Dr. Kadeghe Fue
Academic Year: 2025

Contents

1 1. The Triple Bottom Line **2**

2 2. Economic Impacts (Profit) **2**
2.1 1. Input Reduction (Cost Savings) . . . 2
2.2 2. Yield Maximization (Revenue Increase) . . . 2
2.3 3. ROI Calculation Example . . . 2

3 3. Environmental Impacts (Planet) **3**
3.1 1. Reduced Nutrient Runoff (Eutrophication) . . . 3
3.2 2. Soil Health & Compaction . . . 3
3.3 3. Chemical Load Reduction . . . 3

4 4. Social Impacts (People) **3**
4.1 1. Food Safety & Traceability . . . 3
4.2 2. Labor Safety . . . 3
4.3 3. The Skills Gap . . . 3

5 5. Barriers to Adoption **4**

6 6. Lecture Assessment **5**
6.1 Part A: Standard Concept Questions . . . 5
6.2 Part B: Advanced Analysis (Open Source Data) . . . 5

1 1. The Triple Bottom Line

Technology adoption in agriculture is not driven by coolness; it is driven by value. In the modern era, value is defined by the **Triple Bottom Line (TBL)** framework.

TBL Components

1. **Profit (Economic):** The farm must be financially viable.
2. **Planet (Environmental):** The farming practice must regenerate or sustain the ecosystem.
3. **People (Social):** The technology must benefit the community and workers.

Precision Agriculture (PA) is unique because it is one of the few management strategies that can improve all three simultaneously.

2 2. Economic Impacts (Profit)

Farming is a business. Sensors and robots are capital investments that must generate a Return on Investment (ROI).

2.1 1. Input Reduction (Cost Savings)

The most immediate economic benefit is reducing waste.

- **Overlap Reduction:** Using GPS Auto-Steer reduces overlap by 10-15%. On a 1000-hectare farm, this saves 100-150 hectares worth of fuel, seeds, and chemicals every season.
- **Variable Rate Application (VRA):** Instead of applying 200kg/ha of fertilizer everywhere, VRA applies 0kg/ha in rocky/poor zones.
- **Section Control:** Automatically shutting off sprayer booms at headlands saves 5-10% of chemical costs.

2.2 2. Yield Maximization (Revenue Increase)

PA identifies the "High Potential Zones" of a field. By pushing populations and nutrients higher in these zones (where water/soil can support it), total farm yield increases.

2.3 3. ROI Calculation Example

$$ROI = \frac{(Gain - Cost)}{Cost} \times 100\% \tag{1}$$

Example: A GPS Guidance system costs TZS 10M. It saves TZS 5M per year in fuel/inputs. The payback period is 2 years, which is highly attractive for machinery.

3 3. Environmental Impacts (Planet)

Agriculture is a major contributor to environmental degradation (runoff, greenhouse gases). PA offers engineering solutions to mitigate this.

3.1 1. Reduced Nutrient Runoff (Eutrophication)

When Nitrogen is applied uniformly, low-yielding areas cannot use it all. The excess washes into rivers (e.g., Great Ruaha River), causing algal blooms.

- **PA Solution:** VRA matches N-application to crop uptake, leaving less residue in the soil to wash away.

3.2 2. Soil Health & Compaction

Heavy tractors crush soil pores, reducing water infiltration and root growth.

- **PA Solution: Controlled Traffic Farming (CTF)**. Using RTK GPS to drive on the exact same wheel tracks every year. This leaves 80% of the field uncompacted, increasing water absorption and biology.

3.3 3. Chemical Load Reduction

Herbicides are toxic and expensive.

- **PA Solution: Spot Spraying (See & Spray)**. Robots use cameras to spray *only* the weeds, reducing herbicide volume by up to 90%.

4 4. Social Impacts (People)

4.1 1. Food Safety & Traceability

Consumers want to know where their food comes from. PA systems automatically record every input (chemical type, date, rate).

- **Benefit:** If there is a contamination recall, digital records allow tracking back to the specific meter of the specific field.

4.2 2. Labor Safety

Farming is dangerous (chemicals, fatigue).

- **Automation:** Robots spray toxic chemicals, keeping humans at a safe distance.
- **Auto-Steer:** Reduces operator fatigue, leading to fewer accidents.

4.3 3. The Skills Gap

A negative social impact is the "Digital Divide." PA requires skills in data, electronics, and coding. Rural populations may be left behind if education (like AGE 418) does not catch up.

5 5. Barriers to Adoption

If PA is so good, why doesn't every farmer use it?

1. **Capital Cost:** High upfront cost for sensors and equipment.
2. **Complexity:** Steep learning curve for software.
3. **Interoperability:** Different brands (John Deere vs Case IH) don't always "talk" to each other.
4. **Infrastructure:** Lack of internet (for cloud data) and RTK networks in rural Tanzania.

The Tanzanian Solution: Service Economy

To overcome the Cost barrier, Tanzania is seeing a rise in the **Service Provider Model**. Smallholder farmers do not buy drones; they hire a "Spray Service Provider" (SSP) to spray their 2-acre plot. This spreads the capital cost across hundreds of farmers.

6 6. Lecture Assessment

6.1 Part A: Standard Concept Questions

1. Define the **Triple Bottom Line** in the context of agricultural sustainability.
2. Explain how **Auto-Steer** technology generates direct economic savings for a farmer.
3. What is **Eutrophication**, and how does Variable Rate Nitrogen application help prevent it?
4. Describe the concept of **Controlled Traffic Farming (CTF)**. What environmental problem does it solve?
5. How does **Spot Spraying** (Green-on-Brown) technology impact the "Chemical Load" on the environment?
6. List three **Barriers to Adoption** for Precision Agriculture in developing countries.
7. Explain the **Service Economy Model** (Uber for Tractors). How does it solve the capital cost barrier for smallholders?
8. How does Precision Agriculture contribute to **Food Traceability**?
9. Calculate a simple ROI: A sensor costs TZS 2,000,000. It saves TZS 500,000 per season. What is the payback period in seasons?
10. Why is **Data Privacy** considered a social challenge in digital farming?

6.2 Part B: Advanced Analysis (Open Source Data)

These questions require internet access and the use of open-source datasets/tools.

1. **WaPOR Water Productivity Assessment:** Visit the **FAO WaPOR** portal (https://wapor.apps.fao.org/). *Task:* Locate the **Kamsonga Irrigation Scheme** in Mbarali. Compare the "Gross Biomass Water Productivity" (LWP) of the scheme against the surrounding rainfed agriculture. *Question:* Does the data show that irrigation is being used efficiently (producing more crop per drop), or is water being wasted? How could PA improve this metric?
2. **Environmental Impact Calculation (Fertilizer):** A maize farmer in Iringa uses 200kg/ha of Urea. 40% is leached due to poor timing. *Task:* If VRT reduces leaching to 10%, calculate the reduction in "Wasted N" (kg/ha). Research the "Carbon Footprint of Urea Production" (kg CO2e per kg Urea). How much CO2e is saved per hectare by using VRT?
3. **Soil Erosion Modeling (InVEST/USLE):** Search for the **InVEST** (Integrated Valuation of Ecosystem Services and Tradeoffs) model or the **USLE** (Universal Soil Loss Equation). *Task:* Explain how a high-resolution Digital Elevation Model (from SRTM or Drone) can be used to identify "High Erosion Zones" in the Uluguru Mountains. How would you use a VRT seeder to plant cover crops specifically in these zones?

4. **Economic Feasibility of Drones:** A Drone Service Provider in Morogoro charges TZS 30,000 per acre for spraying. Manual labor costs TZS 15,000 per acre but takes 10x longer and has health risks. *Task:* Create a "Value Proposition" for a farmer. Apart from direct cost, quantify the value of "Speed" (e.g., stopping a Fall Armyworm outbreak in 1 day vs 10 days).

5. **Adoption Heatmap (GIS):** Using **QGIS** and open data (e.g., from **NBS Tanzania** or **WorldPop**), overlay "Cellular Network Coverage" maps with "Cropland Density" maps. *Task:* Identify a district in Tanzania that has High Cropland Density but Low Network Coverage. Why would Cloud-based PA fail here, and why would Edge Computing be the necessary engineering solution?